



Cough and wheezing: asthma and allergic lung diseases

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Conditions with cough and wheezing

Diseases with cough

- Rhinitis, sinusitis, otitis
- Bronchitis (chronic or postviral)
- **Asthma**
- Bronchiectasis
- Cystic fibrosis
- Pneumonia
- Diffuse pulmonary fibrosis

Diseases with airflow obstruction

- Chronic obstructive pulmonary diseases
- **Asthma**
- Bronchiolitis obliterans
- Cystic fibrosis
- Organic or functional laryngeal narrowing
- Tracheal or major bronchial obstruction

Asthma - definition

Asthma is a heterogeneous disease, usually characterized by **chronic airway inflammation**.

It is defined by the **history of respiratory symptoms** such as wheeze, shortness of breath, chest tightness and cough that **vary over time and in intensity**, together with **variable expiratory airflow limitation**.

Classification of asthma

Atopic asthma – in individuals with a tendency to atopy or predisposition to synthesize IgE to common allergens. Usually in children and young adults with a history of infantile eczema and allergic rhinitis → **early-onset**

Nonatopic asthma – in those with no evidence of atopy. Usually in adults → **late-onset**

Conventional

Phenotypes – characteristics or disease traits

Mechanisms – immunological pathways (endotypes) → Type 2-high vs type 2-low

Burden of asthma: global

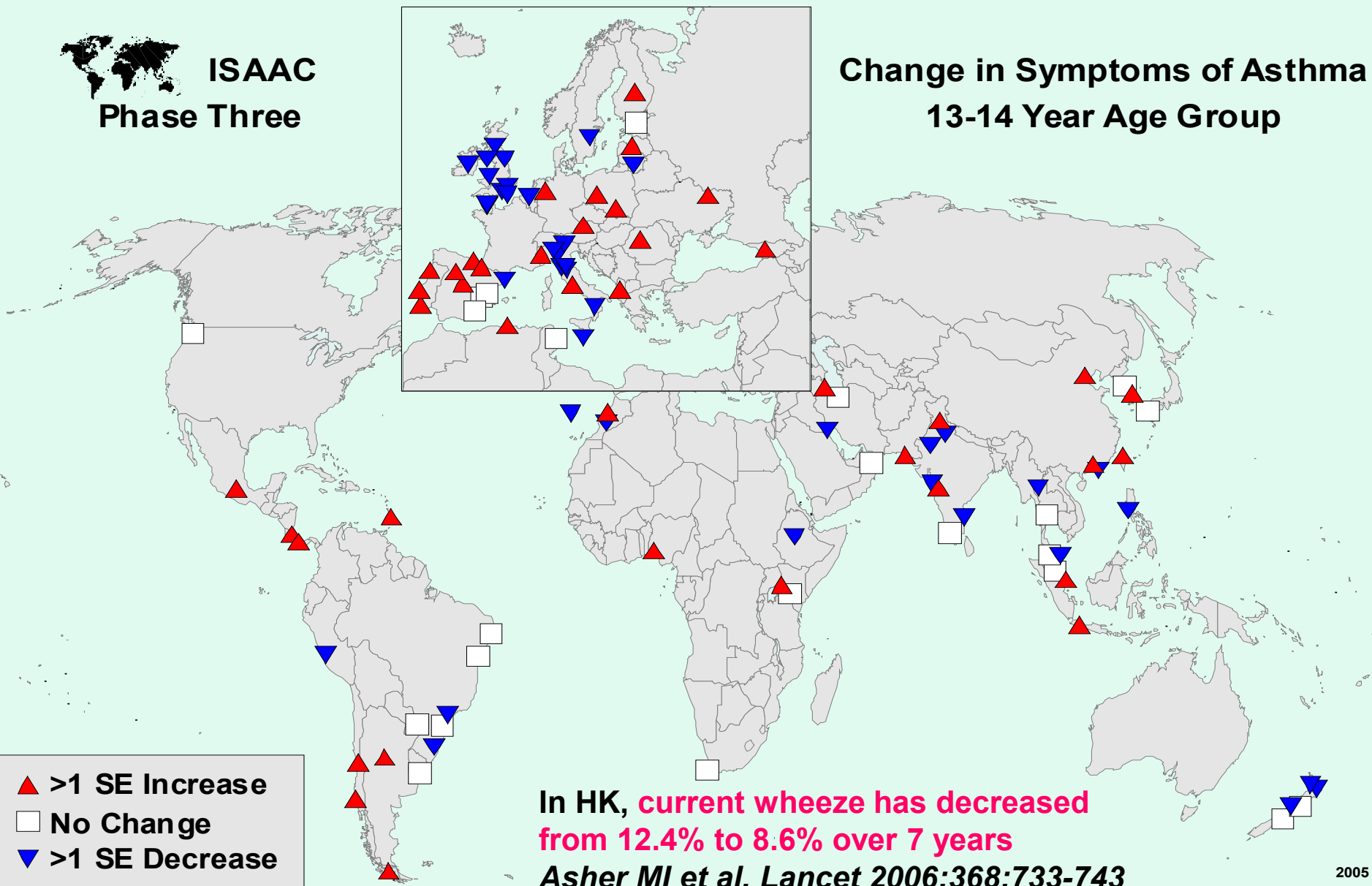
- ~ 300 million affected individuals
- Exact prevalence varies in different populations
 - lack of precise and universally accepted definition of asthma
- Global prevalence 1-18%
- ↓ prevalence in North America and Western Europe
- ↑ asthma symptom prevalence in Africa, Latin America and part of Asia
- 15 million disability-adjusted life years (DALYs) lost annually
- ~ 250,000 deaths from asthma

GINA 2012



ISAAC Phase Three

Change in Symptoms of Asthma 13-14 Year Age Group



Asthma – risk factors

Host predisposition

- Genetics
- Atopy
- Gender
- Obesity

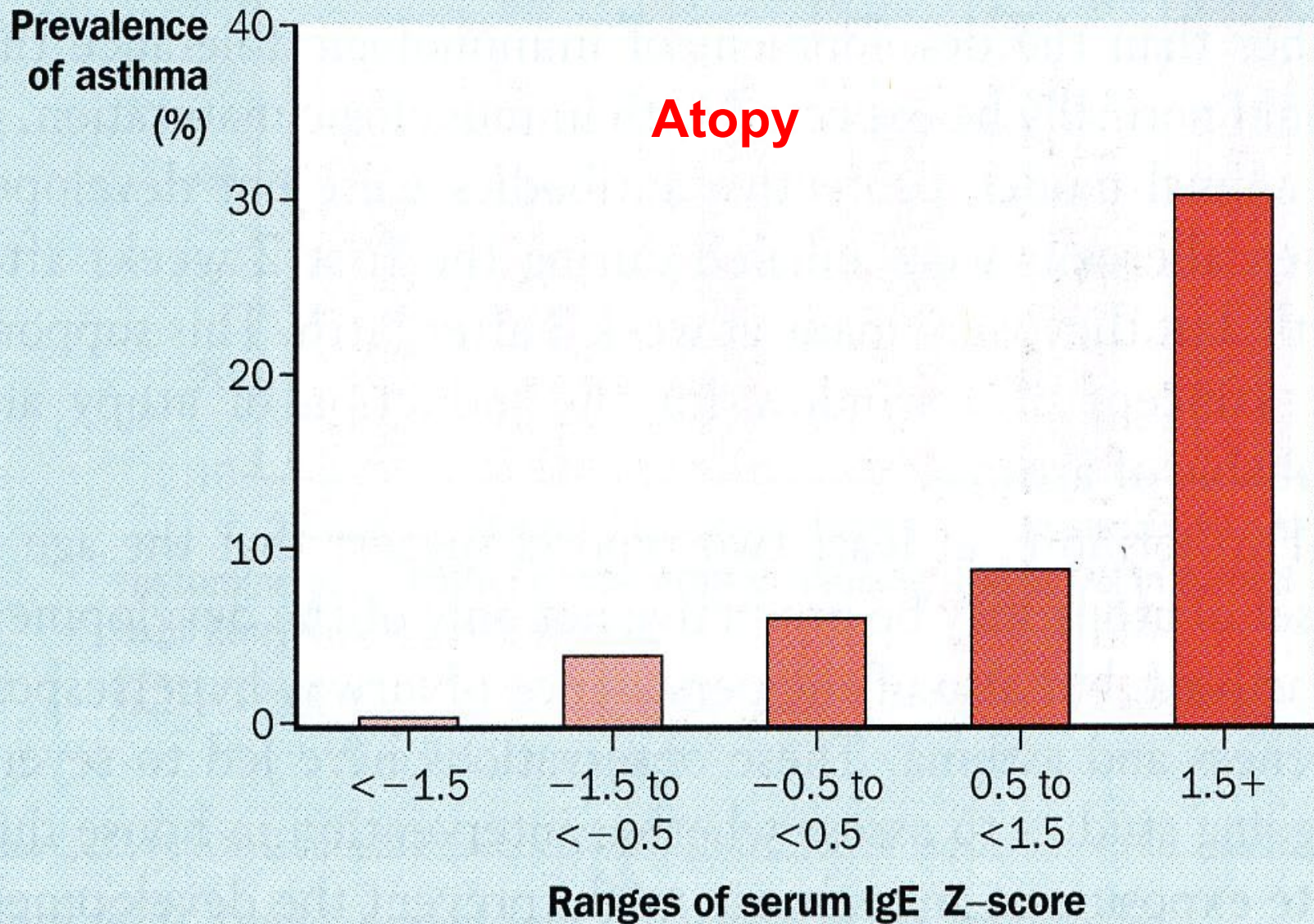
Environmental risk factors

Causes vs Triggers?

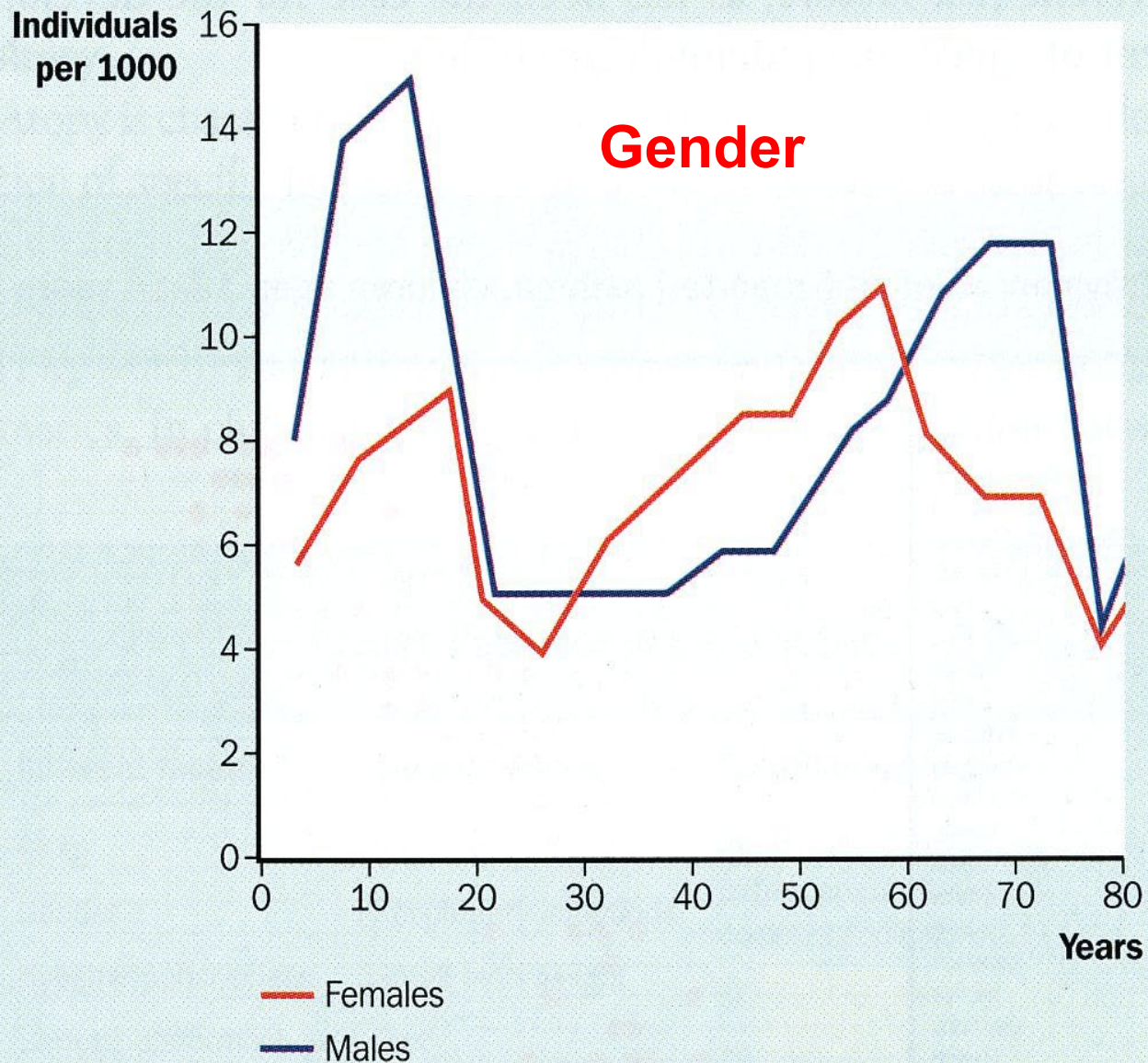
Candidate genes - asthma and atopy

Chromosome	Candidate	Function
5q31	IL-3,-4,-5,-9, -13	Upregulate IgE
5q32	β_2 adrenoreceptors	Bronchodilation
6p	HLA complex	Antigen presentation
6p21.3	TNF α	Inflammatory
11q13	FceR1	Transduction signalling
12q	Interferon γ	Inhibition of Th1
12q	Nitric oxide synthase	
20q	ADAM33	Metalloprotease

Prevalence of asthma in relation to IgE Z-score



Age and sex distribution of asthma



Obesity and asthma

- Asthma more common and difficult to control in grossly obese subjects (BMI > 30 kg/m²)
- Obese asthmatics
 - ↓ lung function
 - ↑ co-morbidities
- Exact mechanisms unknown
 - ? Effect on lung mechanics
 - ? Pro-inflammatory state (cytokines from adipocytes)

Asthma – environmental risk factors

Indoor aeroallergens

- house dust mite
- pets
- cockroaches

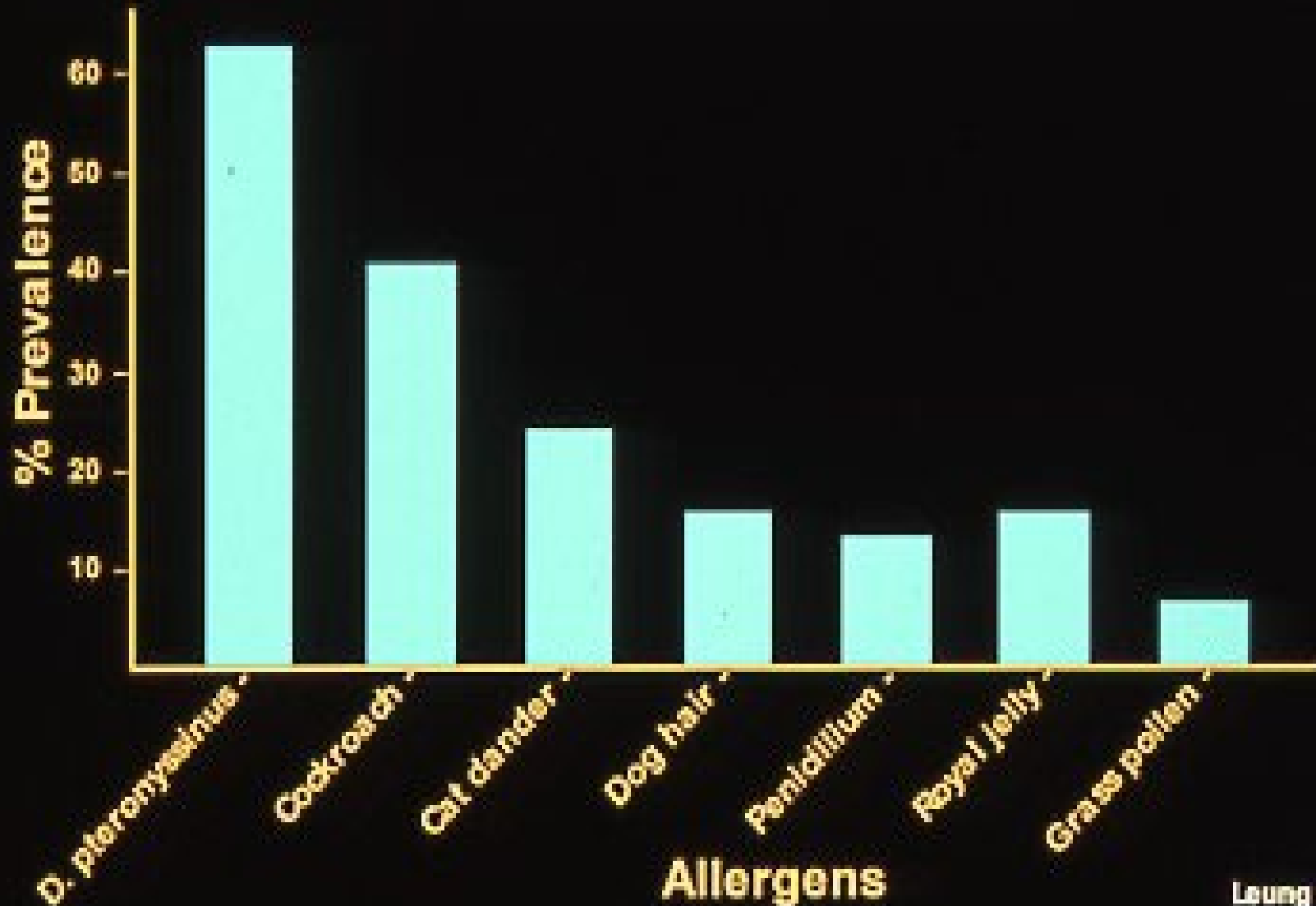
Outdoor aeroallergens

- alternaria

Occupational agents

- Responsible for 5-15% of adult onset asthma

Prevalence of skin test reactivity to common allergens in adult asthmatics in Hong Kong (n= 666)



Leung et al 1998

Asthma - environmental risk factors

Environmental tobacco smoke

Outdoor air pollution

Parental smoking and asthma in children

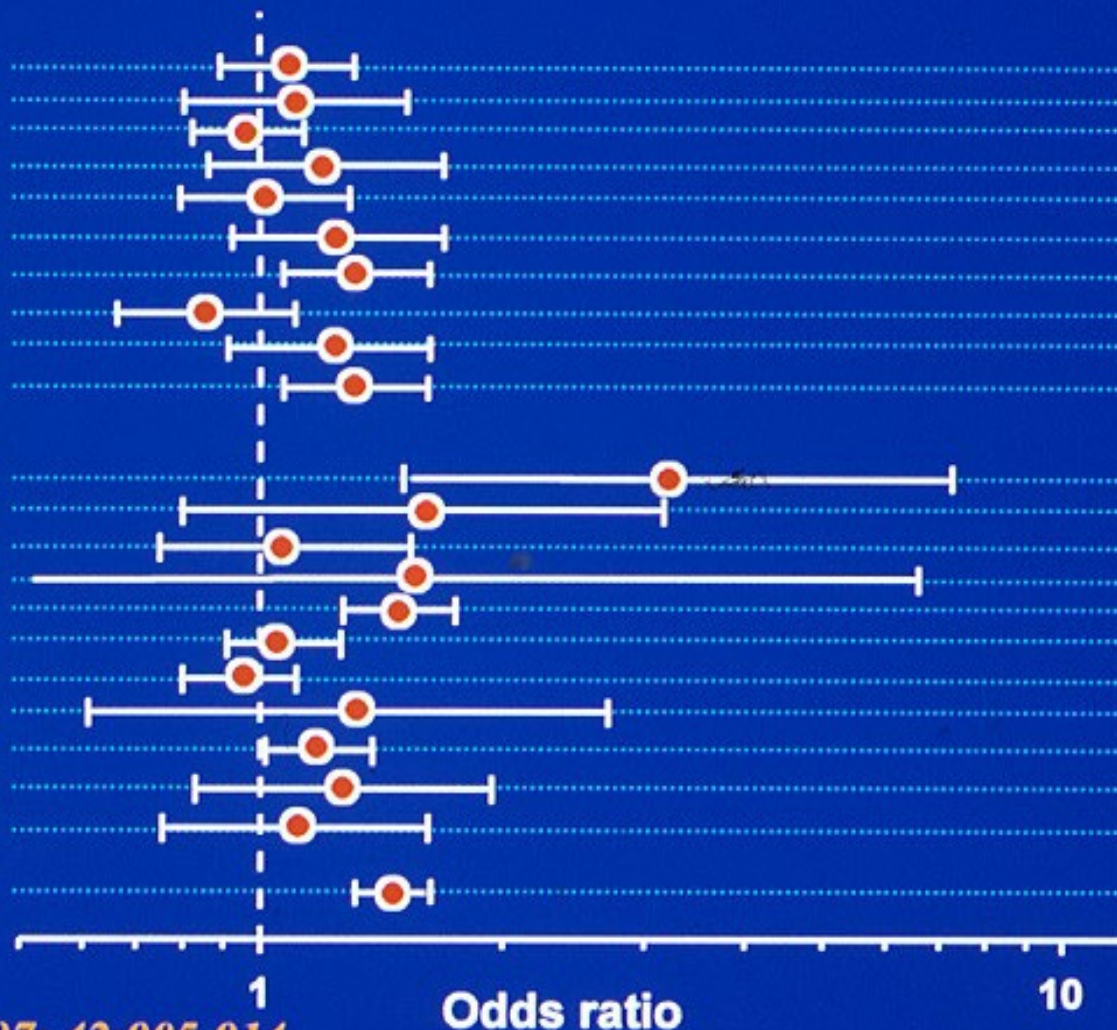
Adjusted ORs

Burchfiel 1986
Somerville 1988
Chinn 1991
Forastiere 1992
Brabin 1994
Kay 1995
Beckett 1996
Peters 1996
Leung 1997
Farber 1997

Unadjusted ORs

Lebowitz 1976
Dodge 1982
Goren 1986
Strachan 1986
Dekker 1991
Moyes 1995
Moyes 1995
Lau 1995
Goren 1995
Bener 1996
Chen 1996

Pooled OR (meta-analysis)



Strachan et al Thorax 1997; 42:905-914

Air pollution and asthma

Increase in health care utilization

- hospital admission
- A and E visits
- unscheduled visits to doctors

Dose-response relationship with air pollutants

Infection and asthma

Viral infection may trigger an acute attack in those with pre-existing asthma

Exposure to some microbes in early infancy may protect against the development of asthma in children (**hygiene hypothesis**)

Hygiene hypothesis

First postulated by Strachan

“Allergic diseases are prevented by infections in early childhood, transmitted by contact with older children”

Microbial exposures associated with less allergy and asthma

Lifestyle (large family size, having older siblings, day care)

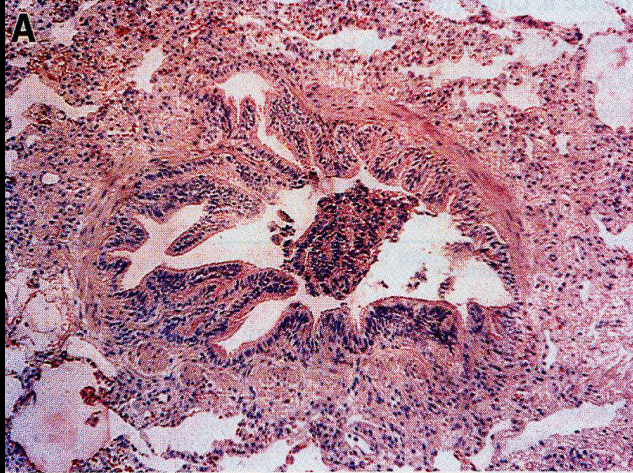
Farming (animal contact, stable exposure, drinking unpasteurized farm milk)

Animal exposure (pet keeping)

Asthma - triggers

Exercise

Cold air

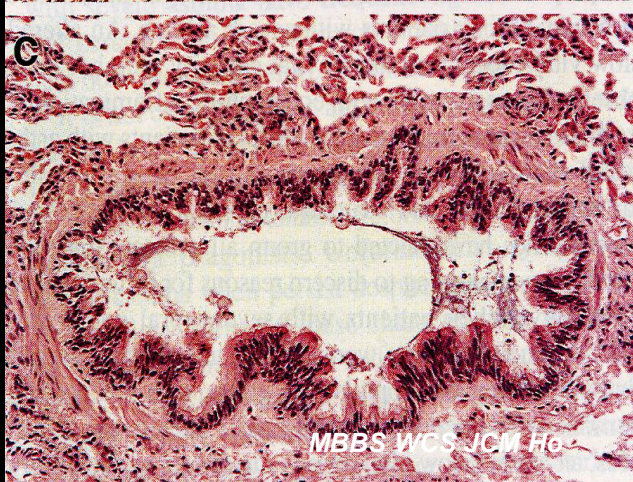
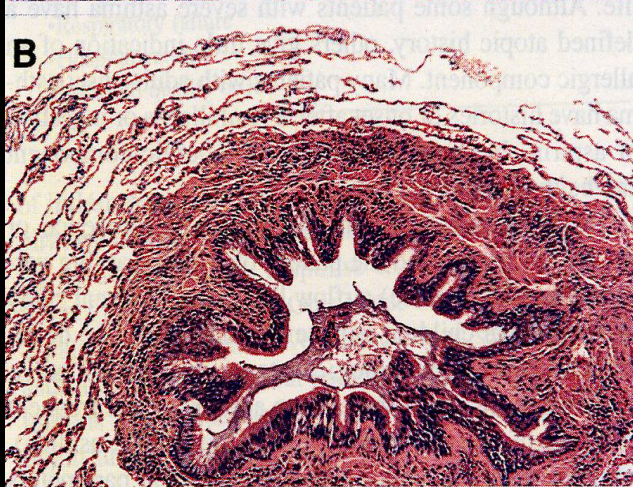


Mechanisms of airflow obstruction in asthma

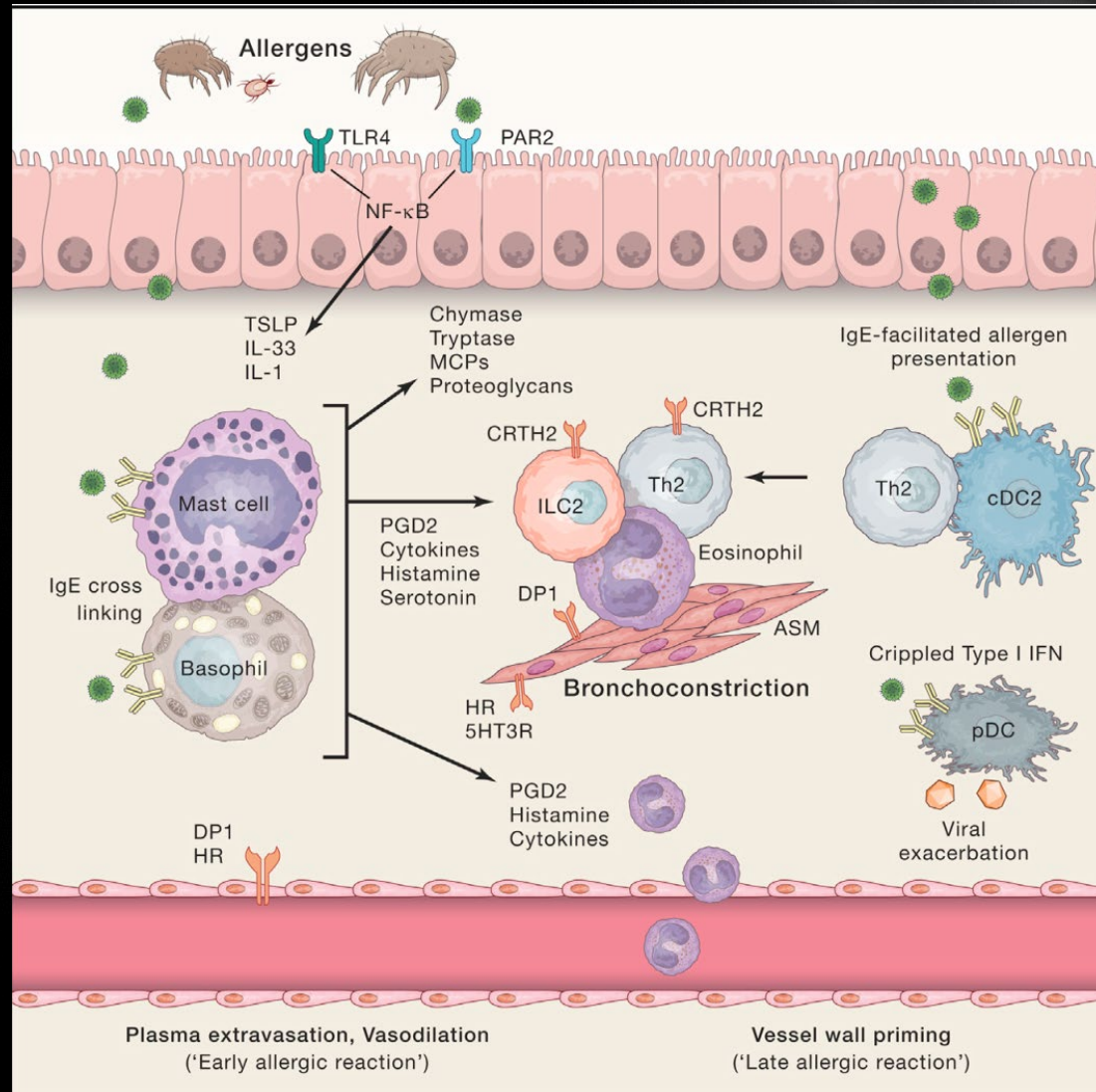
Airway inflammation - eosinophils
lymphocytes

Mucous hypersecretion, goblet cells
hyperplasia

Smooth muscle constriction and
hyperplasia

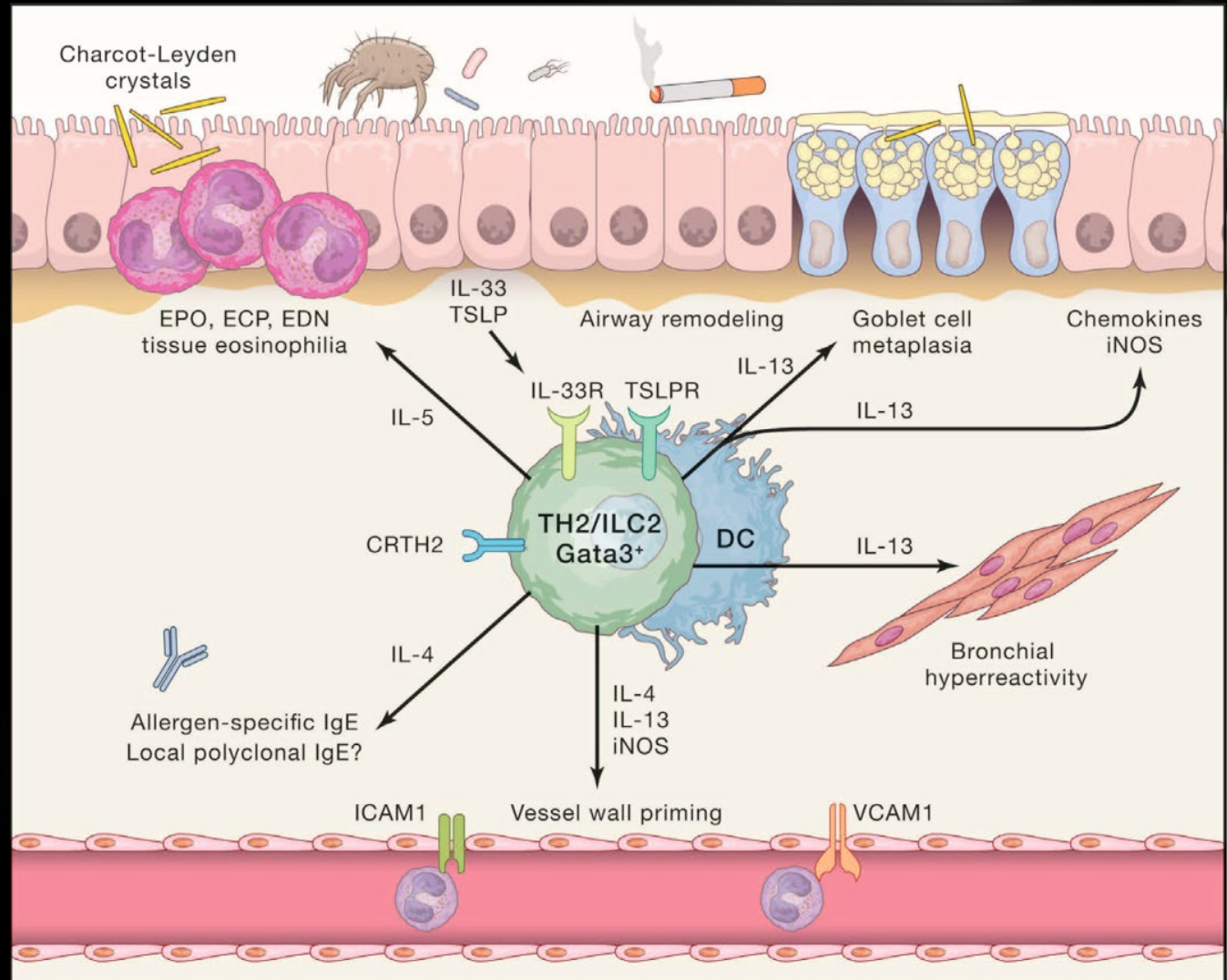


Immunology of asthma: IgE



Immunology of asthma: Th2

Type 2-high
asthma



Asthma- clinical features

Episodes of dyspnea, cough and wheeze

Episodes may be transient or prolonged

Worse at night or early hours of the morning

Some may present as persistent cough

Differential diagnosis of cough and wheeze

Generalized wheeze

- chronic obstructive lung disease
- bronchiectasis
- bronchiolitis obliterans
- viral bronchiolitis (children)

Localized wheeze

- tumour
- foreign body

Asthma - diagnosis

Compatible history

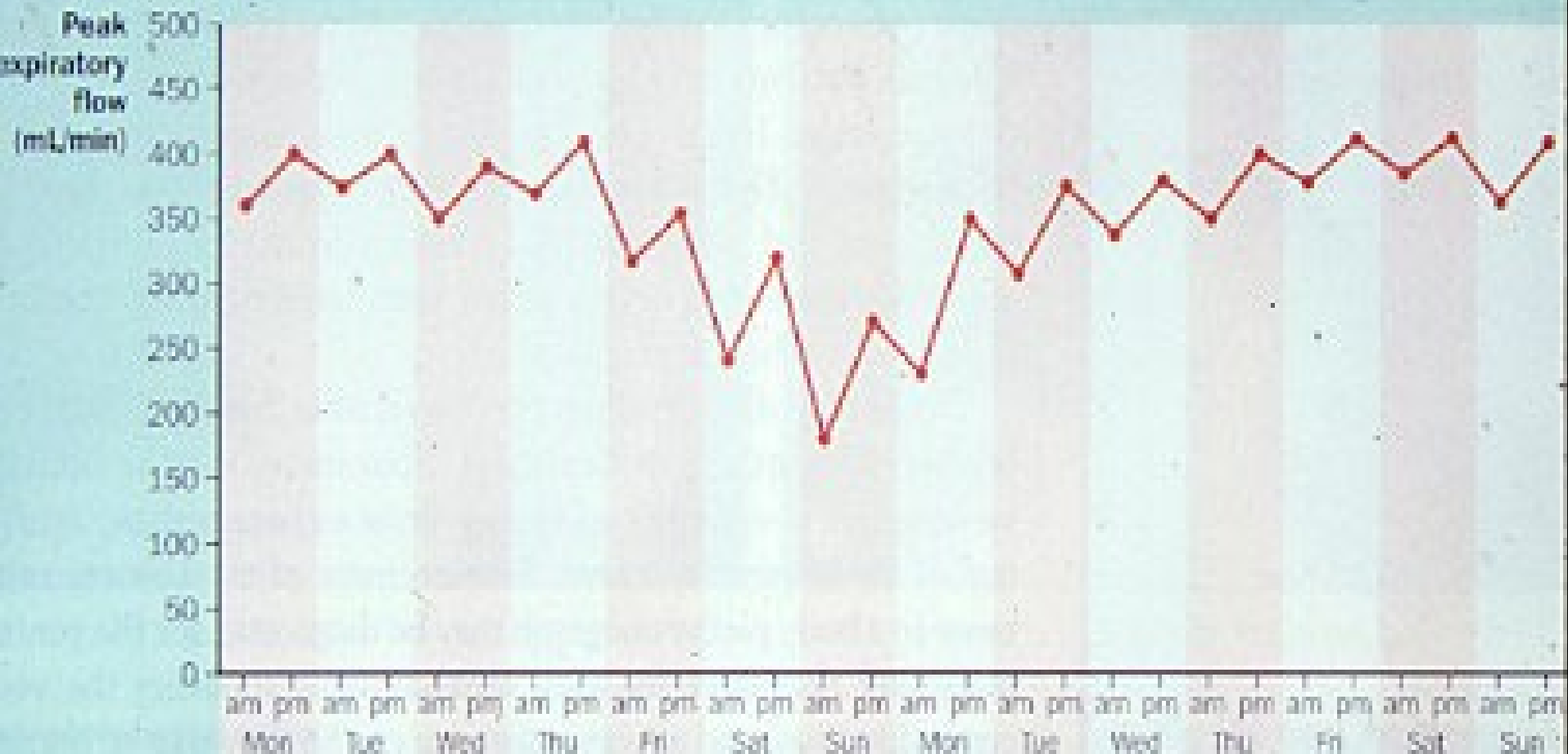
Variable airflow obstruction

- demonstrate presence of airflow obstruction
- improvement in FEV_1 or PEF after bronchodilator
- diurnal variation in PEF or variation in PEF over a period of time

Nonspecific airway hyperresponsiveness →
bronchial challenge test (not essential for dx)



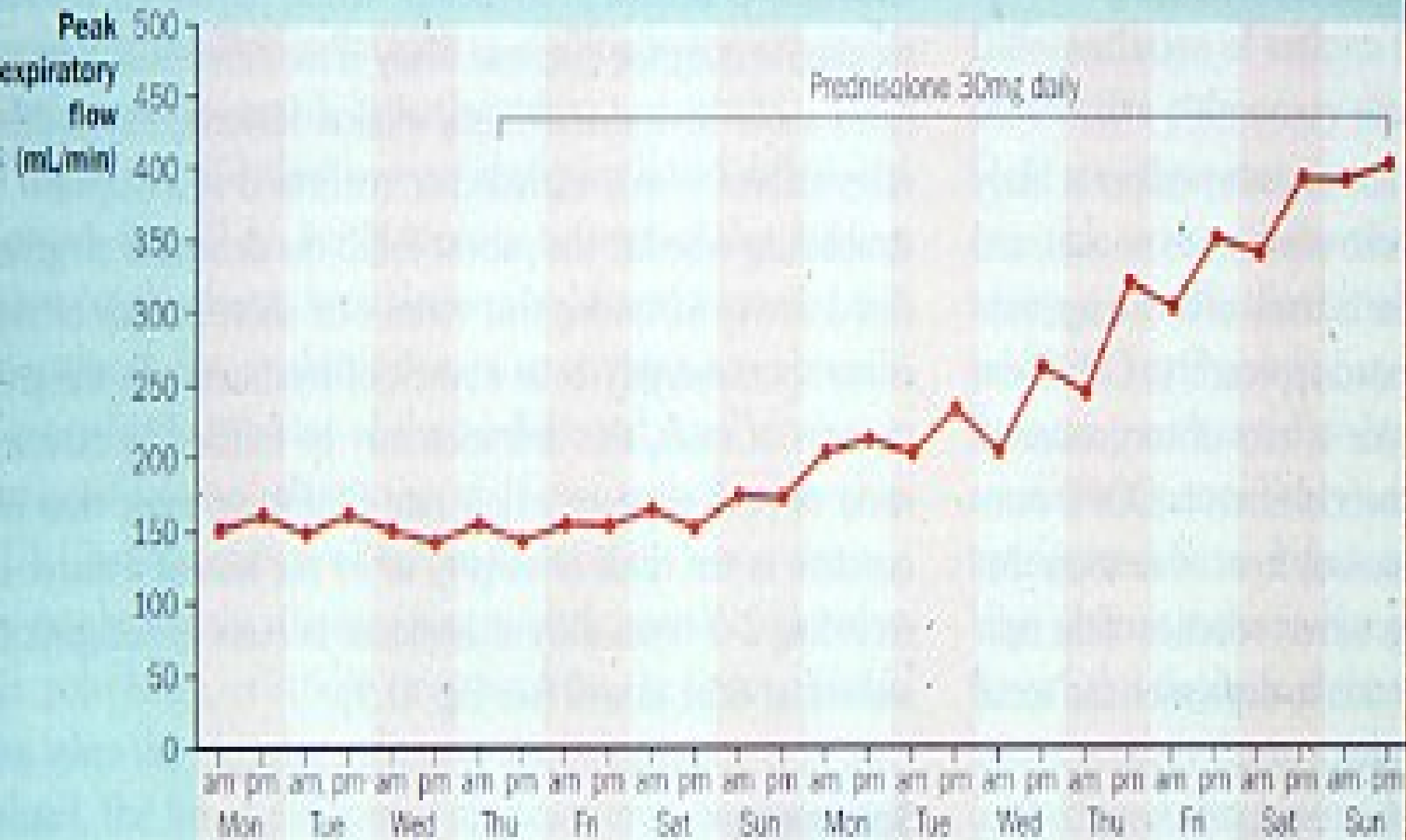
Home peak flow monitoring



Number of doses of reliever medicine taken to relieve symptoms in 24h

Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun
0	0	0	0	1	2	4	3	2	0	0	0	0	0
0	0	1	0	3	4	4	4	2	1	0	0	0	0

Trial of oral corticosteroids



Asthma - assessment

Severity of asthma

- Asthma severity is assessed **retrospectively** from the level of treatment required to control symptoms and exacerbations (GINA 2019)
- Assess asthma severity after patient has been on controller treatment for several months
- Severity is not static – it may change over months or years, or as different treatments become available

Identify triggering factors

Categories of asthma severity

- *Mild asthma*: well-controlled with Steps 1 or 2 (as-needed controller or low dose ICS)
- *Moderate asthma*: well-controlled with Step 3 (low-dose ICS/LABA)
- *Severe asthma*: requires Step 4/5 (moderate or high dose ICS/LABA $\pm$ add-on), or remains uncontrolled despite this treatment

Asthma – treatment objectives

Complete control

No attacks, A & E visits or hospitalization

No symptoms or fewest possible

No limitation of activity

Normal or near normal lung function

Least aerosol bronchodilator use

Least side effects from medications

Asthma - management

Pharmacotherapy

Prevention

Patient education

Two groups of drugs

Relievers

- Bronchodilators

Preventors

- Anti-inflammatory drugs

Bronchodilators - relievers

Short-acting beta-2- agonists

Long acting beta-2-agonists

Xanthines

Anticholinergic

Short-acting beta-2 agonists

Inhaled vs oral

Use when necessary

Salbutamol (Ventolin), terbutaline (Bricanyl)

Side effects

- tremor, headache, arrhythmia
- increase mortality with frequent use alone

Long-acting beta-2 agonists

Prolonged bronchodilation - 12 h

Regular use

Types available

- Salmeterol (Serevent)
- Formoterol (Oxis)

In combination with low dose ICS (<800 ug/day) as an alternative to high dose ICS for asthma control

Anti-inflammatory-preventors

Inhaled or oral steroids

Nedocromil sodium

Sodium cromoglycate

Leukotriene receptor antagonists

Anti-IgE

Other biologics (e.g. anti-IL5)

Corticosteroids

Oral or systemic steroids - acute episodes

Inhaled steroids

- beclomethasone (Becotide, Becloforte)
- budesonide (Pulmicort)
- fluticasone (Flixotide)

Improves symptom control, ↓
exacerbations, ↓ mortality, ↓ lung function
decline

Leukotriene receptor antagonists

Oral, once daily

Used in patients with mild to moderate asthma

Effective in exercise-induced bronchospasm

Effective in preventing aspirin-induced asthma, angioedema etc

Have steroid-sparing effect

Unmask previously undiagnosed Churg-Strauss Syndrome (or eosinophilic granulomatosis with polyangiitis)

Anti-IgE antibody

Omalizumab (anti-IgE Ab)

Indicated in moderate or severe asthma with suboptimal control despite maximal ICS/LABA +/- LTRA or theophylline, or steroid-dependent asthma

Only for atopic asthma (with elevated IgE and positive skin prick test)

Side effects: anaphylaxis

Newer biologics

Anti-IL-5: mepolizumab → targeting eosinophilic asthma

Anti-IL-5R: benralizumab → targeting eosinophilic asthma

Anti-IL-4R α (inhibiting IL-4 and IL-13 pathways): dupilumab → targeting eosinophilic asthma with raised FeNO

→ **Biomarker-selected biologics**

Guidelines for treatment of asthma

British Thoracic Society Guidelines for Asthma management

US National Asthma Education Program

Australian “Asthma Management Plan”

Canadian Asthma Consensus Report

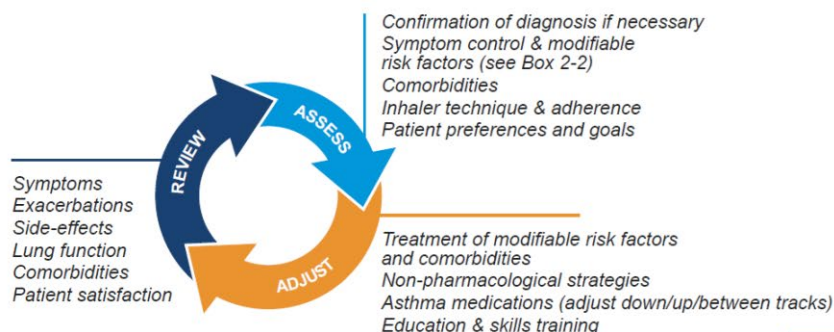
International Union Against Tuberculosis and Lung Disease - Guide for management of asthma in low income countries

Global Initiative for Asthma (GINA)
www.ginasthma.org

GINA 2023 – Adults & adolescents 12+ years

Personalized asthma management

Assess, Adjust, Review
for individual patient needs



Biologics

TRACK 1: PREFERRED CONTROLLER and RELIEVER

Using ICS-formoterol as the reliever* reduces the risk of exacerbations compared with using a SABA reliever, and is a simpler regimen

STEPS 1 – 2

As-needed-only low dose ICS-formoterol

STEP 3

Low dose maintenance ICS-formoterol

STEP 4

Medium dose maintenance ICS-formoterol

STEP 5

Add-on LAMA
Refer for assessment of phenotype. Consider high dose maintenance ICS-formoterol, ± anti-IgE, anti-IL5/5R, anti-IL4Rα, anti-TSLP

RELIEVER: As-needed low-dose ICS-formoterol*

See GINA severe asthma guide

TRACK 2: Alternative CONTROLLER and RELIEVER

Before considering a regimen with SABA reliever, check if the patient is likely to adhere to daily controller treatment

STEP 1

Take ICS whenever SABA taken*

STEP 2

Low dose maintenance ICS

STEP 3

Low dose maintenance ICS-LABA

STEP 4

Medium/high dose maintenance ICS-LABA

STEP 5

Add-on LAMA
Refer for assessment of phenotype. Consider high dose maintenance ICS-LABA, ± anti-IgE, anti-IL5/5R, anti-IL4Rα, anti-TSLP

RELIEVER: as-needed ICS-SABA*, or as-needed SABA

Other controller options (limited indications, or less evidence for efficacy or safety – see text)

	Low dose ICS whenever SABA taken*, or daily LTRA, or add HDM SLIT	Medium dose ICS, or add LTRA, or add HDM SLIT	Add LAMA or LTRA or HDM SLIT, or switch to high dose ICS	Add azithromycin (adults) or LTRA. As last resort consider adding low dose OCS but consider side-effects
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*Anti-inflammatory reliever (AIR)

GINA assessment of asthma control



A. Symptom control

Level of asthma symptom control

In the past 4 weeks, has the patient had:

- Daytime asthma symptoms more than twice a week? Yes ☐ No ☐
- Any night waking due to asthma? Yes ☐ No ☐
- Reliever needed for symptoms* more than twice a week? Yes ☐ No ☐
- Any activity limitation due to asthma? Yes ☐ No ☐

Well-controlled	Partly controlled	Uncontrolled
None of these	1-2 of these	3-4 of these

B. Risk factors for poor asthma outcomes

- Assess risk factors at diagnosis and periodically
- Measure FEV₁ at start of treatment, after 3–6 months of controller treatment to record the patient's personal best, then periodically for ongoing risk assessment

ASSESS PATIENT'S RISKS FOR:

- Exacerbations
- Fixed airflow limitation
- Medication side-effects

House dust mite avoidance

Encasement of mattresses, pillows and duvets with vinyl covers

Hot water wash of all bedding regularly < 2 weeks

Remove carpets

No stuffed toys

Asthma - patient education

What is asthma

What are the triggers and how to avoid them

How the drugs work

How to use inhalers properly

Why is it necessary to use IS regularly

What to do during an acute attack

When to contact a doctor or go to A&E

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)